EFE BATI GÜLEÇ

Materials Engineer/ MSc Mechanical Engineer

Email: efebatigulec@gmail.com, Phone Number:+905078920717 | Address: Buca, Izmir, Turkey

WORK EXPERIENCE

Powder Metallurgy Laboratory, İzmir (Sep 2022 – July 2026)

[Full-time - Material Science Engineer]

Conducted research on Ti-6Al-4V (Ti64) metal matrix composites fabricated via hot pressing powder metallurgy method, investigating the effects of reinforcement content and processing parameters on microstructure and mechanical properties.

Characterized Ti64 metal matrix composites using SEM, XRD, heat treatment and mechanical testing, including tensile and compression tests at room and elevated temperatures.

Investigated the microstructural evolution, mechanical behavior, and hydrogen generation performance of expanded natural graphite (ENG)-reinforced AZ91 magnesium alloy composites for solid-state hydrogen storage applications.

Developed constitutive material models for Inconel 625 using experimental mechanical characterization

Evaluated the mechanical performance and durability of red mud-containing concrete as part of a sustainability-focused project on cement replacement.

Conducted ballistic impact testing on glass specimens using a gas-gun system and analyzed fracture behavior and particle size distribution under varying impact velocities.

Deniz Döküm, Eskisehir (July 2022 – Aug 2022)

[Full-time - Internship]

Contributed to foundry production processes, including mold preparation, casting operations, molten metal pouring, and quality control to ensure product reliability.

Ege Seramik, İzmir (June 2022 – July 2022)

[Full-time - Internship]

Developed and optimized engobe and glaze formulations by conducting experimental studies and material performance evaluations to enhance coating quality and production efficiency.

Özyüzer Research Laboratory, İzmir (Jan 2020 – Apr 2022)

[Part-time - Undergraduate Researcher]

Gained hands-on experience in magnetron sputtering and vacuum systems by depositing and characterizing ITO and ZTO thin films, optimizing process parameters to improve coating quality and functional performance for electronic and optical applications

Teknoma, Izmir (Mar 2021 – Feb 2022)

[Part-time – R&D Engineer]

TUBITAK Research Scholar, Izmir (Jan 2022 – Mar 2022)

[Part-time - Undergraduate Researcher]

Contributed to TÜBİTAK Project 119R038 by depositing, optimizing, and characterizing VO_2 thin films, and evaluating their terahertz (THz) modulation performance for tunable VO_2-based digital metamaterial devices.

EDUCATION

Izmir Institute of Technology, Izmir (2024 – 2026)

Master of Science in Mechanical Engineering

Thesis Name: Microstructural analysis of ceramic-reinforced Ti-6Al-4V metal matrix composites.

Izmir Institute of Technology, Izmir (2018 – 2024)

Bachelor of Science in Material Science and Engineering

Graduation Project: Developed lead-free piezoelectric ceramics for ultrasonic spray nozzle transducer applications through material optimization to enhance performance and environmental compliance.

H. Ahmet Kanatlı and Şehir Anatolian High School, Eskisehir (2014 – 2018)

PUBLICATIONS

- "Microstructure and temperature-dependent tension and compression behavior of Ti-6Al-4V composites reinforced with in-situ formed TiC and Ti_5Si_3 particles via ball-milled SiC whisker addition" (2026). https://doi.org/10.1016/j.msea.2026.151205
- "Effect of strain rate on the compression deformation behavior of SiC whisker/Ti-6Al-4V composites processed by hot pressing" (2026) https://doi.org/10.1016/j.matlet.2026.141394
- "The compaction behavior of Magnesium-Expanded Natural Graphite powder mixtures: Tensile strengths of compacted pellets" (2024). https://doi.org/10.1016/j.matlet.2024.137354
- "Investigation and validation of the flow stress equation and damage model parameters of an electron beam melted Ti6Al4V alloy with a martensitic phase" (2023). https://doi.org/10.1016/j.msea.2023.145590

HARD SKILLS

Computer Programs: SolidWorks, MS Office, KaleidaGraph, X'Pert HighScore Plus, Python, ImageJ.

Technical Knowledge: Powder Metallurgy, Material Characterization, Thin Film Technologies, Vacuum Systems, Titanium and its alloys, Microstructural Characterization, Sintering Furnaces, Recycling of Red Mud, Mechanical Behavior of Materials, Composites

LICENSES & CERTIFICATIONS

TEI 5 Steps Away from Professional Life, TEI Aviation Engines School, Innovation Management & Crisis Management, ISO 9001, Aikido 4th Kyu, Honor Certificate from İZTECH, FMEA (Failure Modes and Effects Analysis)

REFERENCES

Prof. Dr. Mustafa GÜDEN

mustafaguden@iyte.edu.tr, +90 533 304 5463

Asst. Prof. Dr. Kemal DAVUT

kemaldavut@iyte.edu.tr, +90 533 259 3116

Prof. Dr. Umut ADEM

umutadem@iyte.edu.tr, +90 538 865 3085